VIT

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:

Continuous Assessment Test (CAT) – II - MARCH 2026

Programme	:	B. Tech (CSE)	Semester	:	Fall 25-26
Course Code & Course Title	:	BCSE301L & Software Engineering	Class Number	:	CH2025260502037 CH2025260502033 CH2025260502026
Faculty	:	Dr. R. PRABHAKARAN Dr. ALOK CHAUHAN Prof. S. BERIN SHALU	Slot	:	D1+TD1
Duration	:	90 Minutes	Max. Mark		50

General Instructions:

- Write only your registration number on the question paper in the box provided, and do not write other information
- Only a non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1		A private hospital plans to develop a Hospital Management System (HMS). The proposed system should allow doctors to view patient records, enable receptionists to schedule appointments, generate bills automatically, store patient medical history securely, provide 24/7 availability, and respond within three seconds during peak hours. Based on the given scenario, answer the following questions:			
	(i)	Identify and classify the given requirements into functional requirements, non-functional requirements, and domain requirements. (5 Marks)	10	CO3	K3
	(ii)	Which of the stated requirements can be considered quality attributes? Justify your answer. (2 Marks)			
	(iii)	Identify any ambiguous requirement in the scenario and rewrite it in clear and measurable form. (3 Marks)			

2		In a legacy inventory management module (written in an object-oriented language), you find a class called InventoryManager with the following responsibilities and dependencies: • Loads product data from a CSV file on disk. • Validates product codes and quantities using hardcoded regex and business rules. • Calculates reorder levels based on sales history (fetched via direct SQL queries to the database). • Sends low-stock email alerts using a concrete SmtpEmailSender class. • Logs every action by writing directly to a fixed log file path. • Updates stock in an external warehouse system via a REST API call (using hardcoded URL and credentials). • Checks user permissions before allowing stock adjustments (using a global Session object). • Formats and prints a stock report to console. (i) Identify primary type of cohesion exhibited by the InventoryManager class. Breakdown this class into two new cohesive modules, suggest a reasonable division (name the two new classes and list which responsibilities go into each). (5 marks). (ii) Illustrate four different types of coupling present in InventoryManager class with the help of concrete examples from the given scenario. Explain why this coupling makes maintenance or testing more difficult. (5 marks)	10	CO3	K4
3		Develop a modern Court Case Management System (CCMS) to replace the largely paper based processes currently used in many courts, which frequently result in misplaced files, delayed hearings, and poor transparency for all parties involved. The new web based system aims to fully digitize key workflows, including case registration, hearing scheduling, document submission and management, judgment recording, and publication of orders or judgments, while also supporting appeals tracking. The system defines several key user roles with strict role-based access control (RBAC): • Lawyers can register new cases, upload supporting documents, view case details, track upcoming hearing dates, and receive automated notifications. • Judges can view their assigned case calendars, review uploaded documents and evidence, schedule or reschedule hearings, record court proceedings and minutes, enter and publish judgments or orders. • Court staff / Clerks can manage overall case progression, update status, generate and send notifications to parties, handle administrative tasks related to case flow. • Citizens / Litigants (public users) can view the public status of cases they are involved in, receive status update notifications, and access published judgments or orders.	10	CO4	K3

		• Administrators have oversight privileges to manage user accounts, configure system rules, and monitor audit logs. Critical non-functional requirements include: strong confidentiality and data privacy, comprehensive audit trails for all significant actions (who did what, when), compliance with relevant legal data protection regulations, and reliable notification mechanisms. (i) Explain how the principles of encapsulation, inheritance, polymorphism, and abstraction can be applied in designing the Court Case Management System. (5 Marks) (ii) Design a UML class diagram that captures the core domain entities (such as Case, Hearing, Document, Judgment, User/Actor, Party, Notification, etc.), their key attributes, relationships (associations, compositions, inheritances, multiplicities), and important methods/operations relevant to the system's behavior. (5 Marks)			

4		void grade(int marks)
		{
		if (marks >= 90)
		printf("A");
		else if (marks >= 75)
		printf("B");
		else if (marks >= 50)
		printf("C");
		else
		printf("F");
		}
		(i) Design flow graph for the given code fragment (2 Marks)
		(ii) Find the number of independent path and cyclomatic complexity in the flow graph (2 Marks)
		(iii) A bank ATM allows customers to withdraw cash with the following constraints:
		• Minimum withdrawal amount: 100
		• Maximum withdrawal amount: 10,000
			10	CO5	K5

		• Withdrawal amount must be a multiple of 100 Use Boundary Value Analysis to create test cases for this system. (3 Marks)			
	(iv)	A social media app only allows users who are 13 years or older up to 85 to register. Using Equivalence Partitioning, identify the valid and invalid input partitions for the age input and design corresponding test cases. (3 Marks)			
5		An online e-commerce platform regularly introduces new features such as payment gateways, discount systems, and product recommendation engines. After several updates, previously stable modules started failing, affecting the reliability of the system.	10	CO4	K5
	(i)	Analyze the importance of Regression Testing in maintaining system stability when software evolves. (3 Marks)			
	(ii)	Illustrate the regression testing process and discuss different regression testing strategies such as retest-all, regression test selection, and test case prioritization. (4 Marks)			
	(iii)	Evaluate which strategy would be most effective for large-scale systems and justify your answer.(3 Marks)			

**********All the best **********